



EXOTOPIA

/ SCD Hub

What if every community on Earth had a place in the
universe —
and a way to [prove](#) the work they do to protect it?

The work is real. The credit isn't.

Community groups around the world — watershed councils, youth science groups, smallholder farmers — are doing irreplaceable environmental monitoring. Water quality. Soil health. Species counts. Tick surveillance.

Most of this work goes unrecognized, unpaid, and disconnected. A school in Vermont and a youth group in Kenya may be running identical stream health protocols. They have no idea the other exists. Neither receives a credential for the work. Neither can easily share what they've learned.

SCD Hub / Exotopia is a platform designed to fix all three at once:



The platform is **open source**, uses **open data standards** (GBIF, EPA ECHO, iNaturalist), and routes income directly to the people doing the work — not to a platform middleman.

Who's already using it: Community groups in Lamu County (Kenya), Vermont river monitoring networks, Costa Rica biodiversity projects, and Front Range Colorado citizen science groups. The tools work offline — no data plan required in the field.

Real communities. Real work.

A SAMPLE OF WHO'S IN THE NETWORK

MPEKETONI, LAMU COUNTY · KENYA

Uni-Kibaoni Peace Youth SHG

Led by Muirithi Jariffe, this self-help group monitors water quality and food security along the Lamu coastal plain. Five years of BMWP macroinvertebrate data, soil health records, and eco:certificate credits for monitoring work that previously went unacknowledged.

VERMONT · NORTHEAST USA

Freshwater Phenology Network participants

Schools and watershed councils running Secchi disk, dissolved oxygen, and macroinvertebrate monitoring on local streams and lakes. Data feeds into USA-NPN's national phenology database. Curriculum units align with Vermont Agency of Education STEM frameworks.

FRONT RANGE · COLORADO

Boulder County citizen science groups

You. CoCoRaHS observers. Boulder Waterkeeper stream monitors. CU Boulder field ecology students. Rocky Flats neighborhood PFAS watchers. Open Space trail monitors. The platform's protocols are designed to make your existing data globally legible.

"The monitoring a Vermont school does for their local stream is directly comparable to what Uni-Kibaoni does for their river. That's by design — we use the same open protocols so the data connects."

An exoplanet settlement is a strange home for a watershed council. It works anyway.

Four things the platform provides, one visual metaphor holding them together



Eco:Certificates

Verifiable digital records of real field work. Each certificate is an open badge anchored on Algorand. No wallet required to earn one.



Bounty System

Specific monitoring tasks, posted publicly, claimed by community members. Payment via M-Pesa, bank transfer, or Algorand. Groups can post their own bounties.



Training Library

Video curriculum for ecological monitoring, vocational skills (carpentry, wiring, EMT, accounting), and citizen science protocols. Works offline.



Settlement Map

Each community gets a navigable "exoplanet settlement" — a visual home for their data, achievements, and identity in the global network.



Why exoplanets? The settlement is a persistent, visual identity for a real community. It holds their monitoring records, certificates, and connections. It's also a useful fiction: communities 300 km apart can be "neighbors" on the same planet if they're working on the same ecological problem.

BOULDER COUNTY

You're already doing this work.

Boulder County has some of the densest citizen science infrastructure in the country. NCAR. NOAA. CoCoRaHS (headquartered in Fort Collins). CU's ecological monitoring programmes. The Boulder Waterkeeper. Open Space trail counters. What the platform adds is *connection* — your data talking to a global network, and global data talking back.

- **Boulder Creek / St. Vrain monitoring** — the platform's BMWP macroinvertebrate protocols and Secchi disk methods are identical to what Colorado's stream monitoring programmes already use. Existing records can be imported.
- **Front Range tick ecology** — our Tick Watch Northeast curriculum applies directly to Rocky Mountain wood tick and western black-legged tick surveillance. We're actively looking for Colorado SMEs to validate and extend it.
- **CU Boulder / NCAR connections** — the platform's phenology protocols feed directly into USA-NPN's National Phenology Database, which NCAR uses for seasonal forecasting models. Student project data can contribute.
- **Vocational training** — our curriculum library covers carpentry, electrical wiring, EMT response, and business accounting. Workforce development orgs in Boulder County: get in touch about co-curating Front Range relevant content.

▲ PFAS / PFOS — TWO ACTIVE FRONTS IN BOULDER COUNTY

ROCKY FLATS LEGACY SITE

Groundwater contamination persists for Louisville, Superior & N. Jefferson County. EPA Method 533 chain-of-custody guides and ECHO submission pathways are in the bounty system — free to use.

NEDERLAND - ACTIVE COLLABORATION

Working with Cat (Town of Nederland sustainability manager) on zero waste education + a PFAS testing regime for the waste transfer site's **Greener Mountain EarthFlow** 20-yd in-vessel composter.

▲ GARDEN CONTAMINATION PATHWAY — DIRECT HUMAN EXPOSURE RISK

- PFAS moves from amended soil into plant tissue. **Leafy greens accumulate highest:** lettuce, spinach, kale, chard. Root vegetables (carrots, beets) and fruiting crops (tomatoes, squash) take up less — but still measurable.
- In-vessel systems run at 55–70°C. **PFAS are thermally stable** — composting **does not destroy them**. It concentrates them: less leaching during processing means more retained in finished product.

- Short-chain PFAS (PFBS, PFHxA) are most mobile in soil and most readily taken up by roots. Long-chain PFOS/PFOA bind to soil organic matter — but still transfer. **No chain length is safe to ignore in a food garden.**
- Colorado **HB 23-1034** (2023) restricts PFAS in food packaging 2024–2026 — this reduces future inputs. But existing feedstock contamination requires testing *now*, before compost is distributed.

TESTING + MITIGATION STEPS FOR NEDERLAND

1. Test finished compost before any distribution to food gardens — **ASTM D8421**

2. Establish hold threshold: Maine's agricultural trigger is 2.5 ppb total PFAS. If

FOR THE CURIOUS

The math behind the metaphor is as real as it gets.

A glimpse at the framework — full writeup at exotopia.org/blog

Every community in the network has a profile across **eight capability dimensions** — the same eight that appear in our human rights and Rights of Nature framework. These eight dimensions map onto a mathematical structure called **E8**, one of the most symmetric objects in mathematics. "Near" communities in E8 space are working on analogous problems and have things to teach each other.

x ₁	Land stewardship	x ₂	Water & watershed
x ₃	Biodiversity	x ₄	Community governance
x ₅	Technology access	x ₆	Educational reach
x ₇	Economic inclusion	x ₈	Climate adaptation

THE LEECH LATTICE (Λ_{24})

Three stacked E8 blocks form a 24-dimensional structure called the **Leech lattice**. A community's full address in this space encodes: what it currently achieves (Block 1), what its ecosystems need (Block 2), and how it relates to time and network (Block 3). Parallel "universe" simulations for what-if scenarios are different addresses in the same 24D space.



Field work certificates use **PLONK/halo2 zero-knowledge proofs** — the same cryptographic infrastructure used in Ethereum's rollup ecosystem. No trusted setup. Certificates are verifiable by anyone with the public inputs, without revealing the field worker's identity. E8 root vectors serve as tamper-resistant identity slots.

Four ways to plug in from **Boulder**.

No blockchain expertise required. No app to install. No fee.

- 1 Monitor** **FIELD WORK**
Use our open protocols (BMWP, Secchi disk, PFAS sampling, phenology) for monitoring work you're already doing. Your data earns eco:certificate credits and connects to the global dataset. Works offline on any Android or iPhone.
- 2 Teach** **EDUCATORS**
Run one of our citizen science curriculum units — Tick Watch Northeast, Cyanobacteria Watch, or Freshwater Phenology Network — in your classroom, youth group, or community workshop. Full lesson plans, field protocols, and data submission guides. Free to use, adapt, and share.
- 3 Build** **DEVELOPERS**
The platform is open source (GPL v3). We need Vue/Quasar developers, Supabase/PostGIS contributors, GIS specialists, and anyone who knows how to wire a PLONK proof to a front-end. GitHub: [biomassives/vercel-html-exotopia.org](https://github.com/biomassives/vercel-html-exotopia.org)
- 4 Connect** **COMMUNITY ORGS**
If you coordinate a watershed group, school science programme, community garden, or vocational training org — ask about becoming a **group coordinator**. Post bounties for your community's monitoring priorities. Co-curate curriculum for Front Range conditions.

EMAIL acmeideal@gmail.com

WEB exotopia.org

CODE github.com/biomassives/vercel-html-exotopia.org

exotopia.org

A navigable universe for community science — where every settlement on an exoplanet maps to real people doing real work on Earth.

24

Leech lattice dimensions mapping community capability + nature rights + cultural memory

8

E8 capability axes — land, water, biodiversity, governance, technology, education, economy, climate

GPL3

Licence. Every line of code is open source. Fork it, run your own instance, adapt for your region.

0%

Platform cut from community monitoring payments. Income goes directly to field workers and SMEs.

Made in Boulder & Nairobi & Mpeketoni
& wherever someone is monitoring
something that matters.

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